

```
In [1]: import read_mist_models
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: eep1 = read_mist_models.EEP('MIST_v1.2_feh_p0.00_afe_p0.0_vvcrit0.0_EEPS/00100M.track.eep')
eep5 = read_mist_models.EEP('MIST_v1.2_feh_p0.00_afe_p0.0_vvcrit0.0_EEPS/00500M.track.eep')
eep10 = read_mist_models.EEP('MIST_v1.2_feh_p0.00_afe_p0.0_vvcrit0.0_EEPS/01000M.track.eep')
eep20 = read_mist_models.EEP('MIST_v1.2_feh_p0.00_afe_p0.0_vvcrit0.0_EEPS/02000M.track.eep')
```

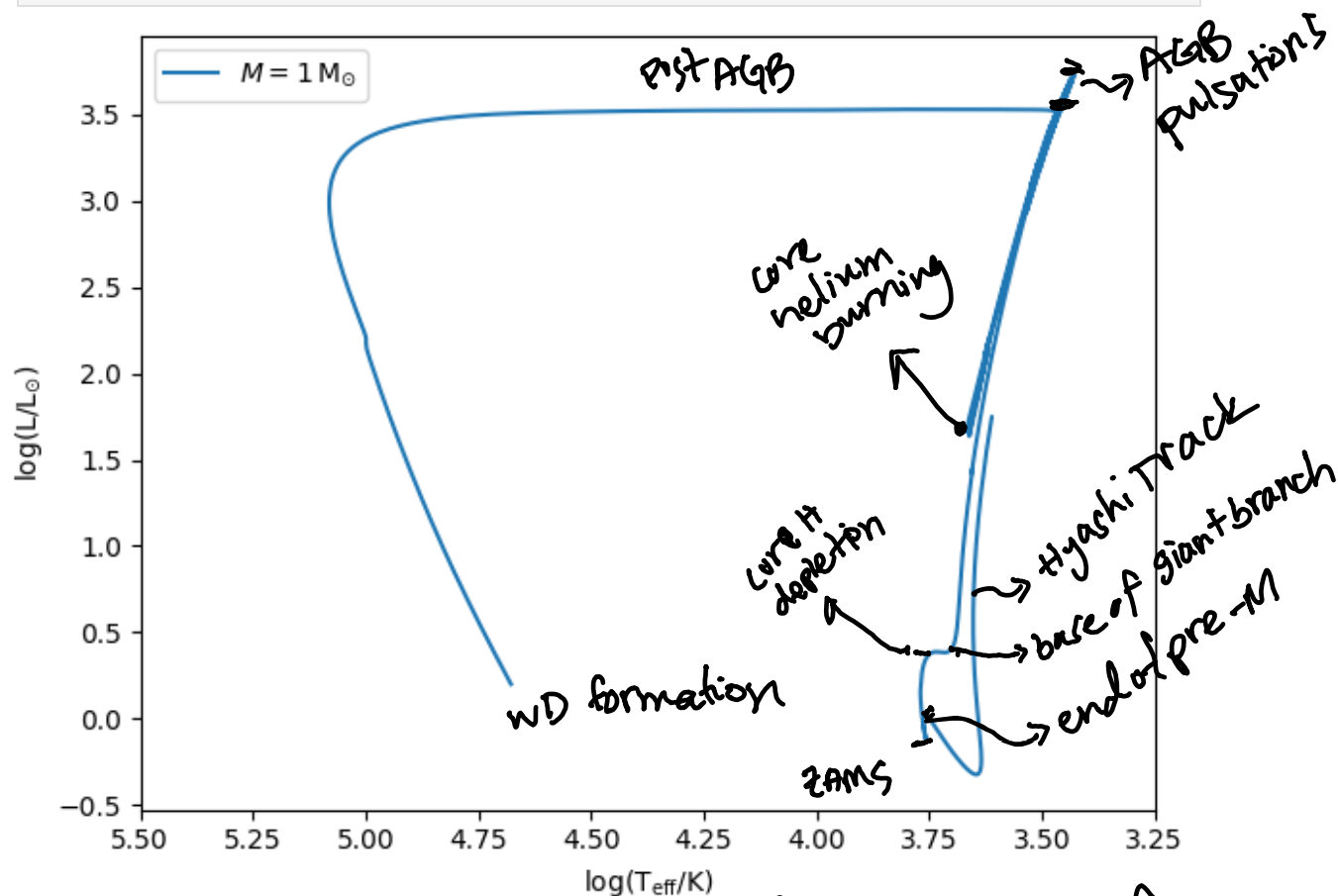
Reading in: MIST_v1.2_feh_p0.00_afe_p0.0_vvcrit0.0_EEPS/00100M.track.eep

Reading in: MIST_v1.2_feh_p0.00_afe_p0.0_vvcrit0.0_EEPS/00500M.track.eep

Reading in: MIST_v1.2_feh_p0.00_afe_p0.0_vvcrit0.0_EEPS/01000M.track.eep

Reading in: MIST_v1.2_feh_p0.00_afe_p0.0_vvcrit0.0_EEPS/02000M.track.eep

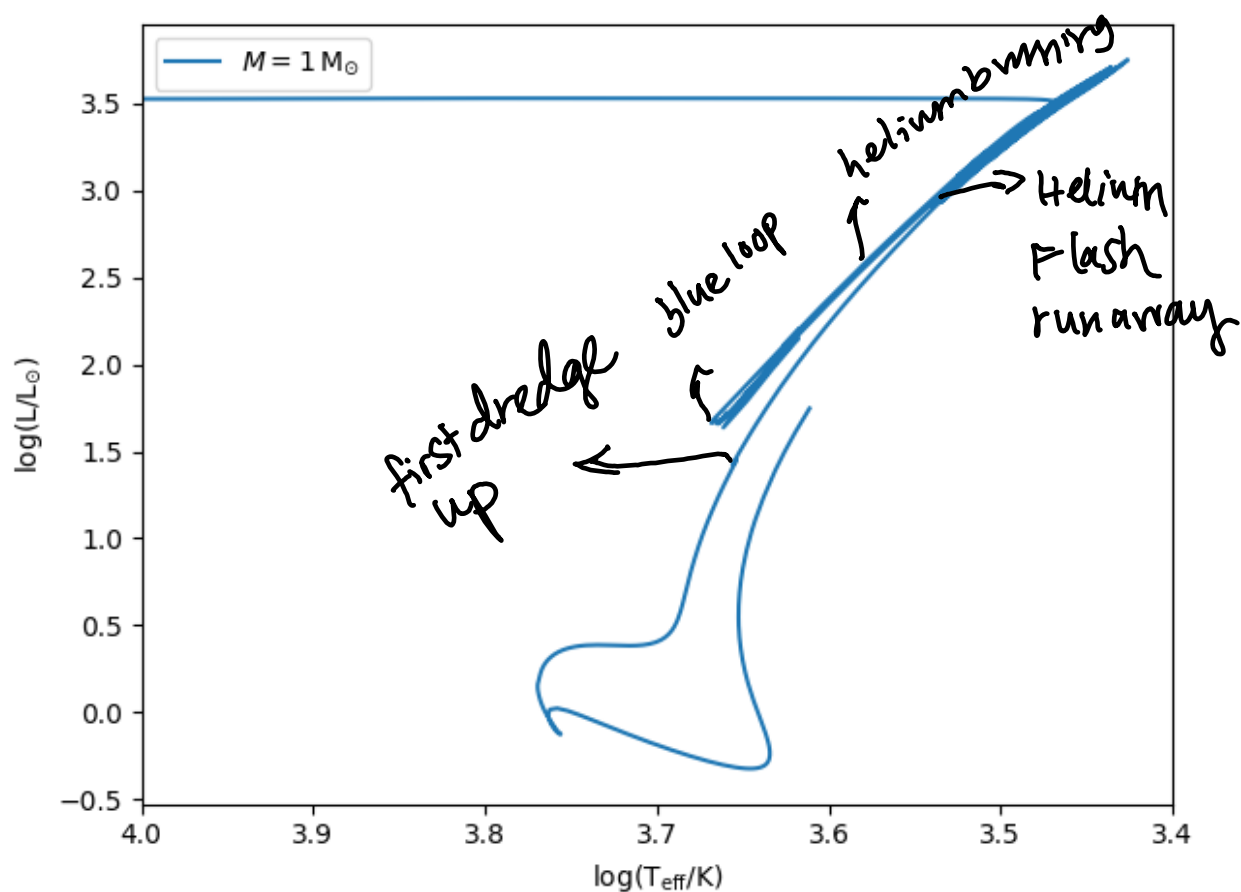
```
In [3]: plt.plot(eep1.eeps['log_Teff'], eep1.eeps['log_L'], label=r'$M=1\, \rm{M}_{\odot}$')
plt.xlim(5.5, 3.25)
plt.xlabel(r'$\log(T_{\rm{eff}}/K)$')
plt.ylabel(r'$\log(L/L_{\odot})$')
plt.legend()
plt.tight_layout()
```



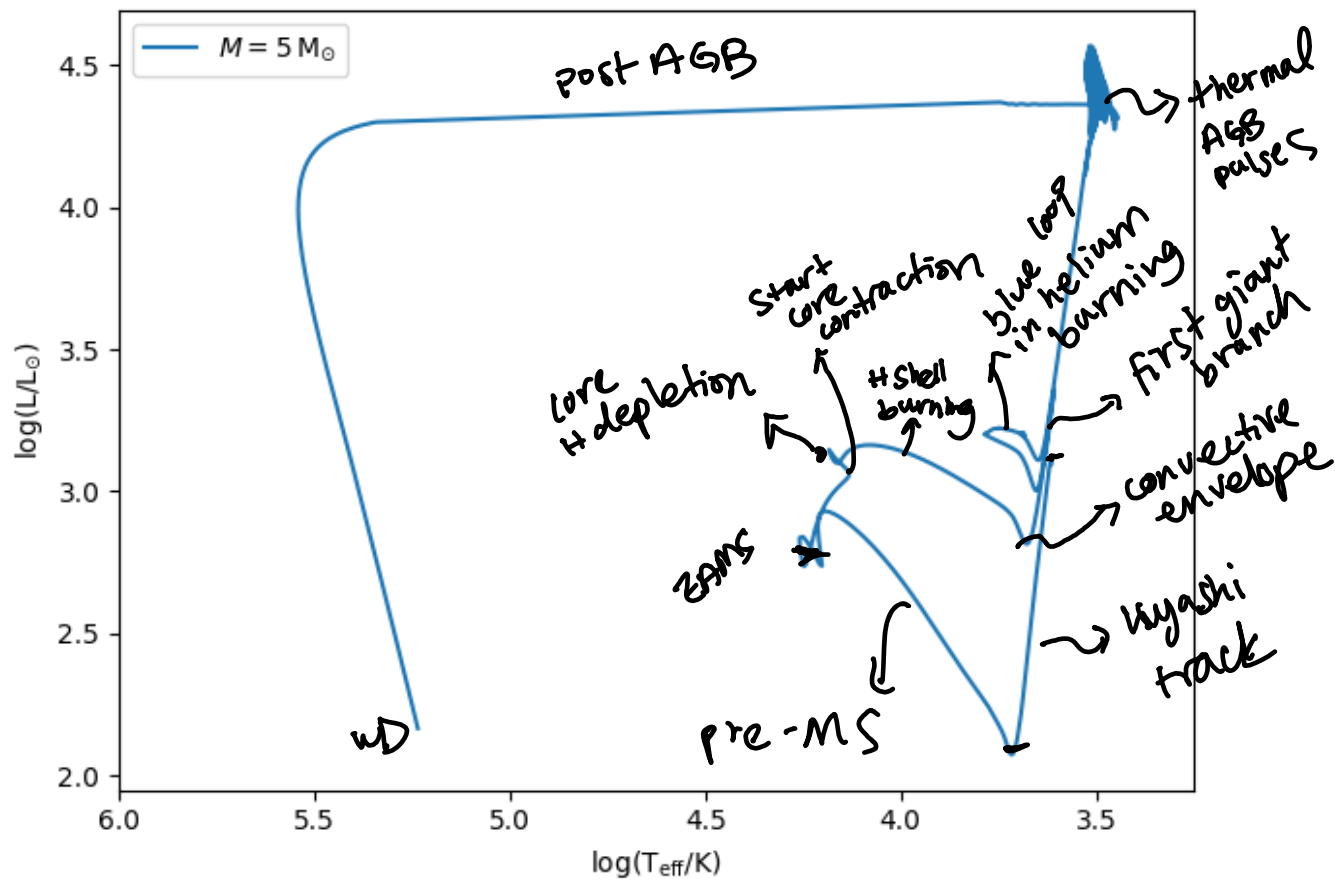
```
In [4]: plt.plot(eep1.eeps['log_Teff'], eep1.eeps['log_L'], label=r'$M=1\, \rm{M}_{\odot}$')
plt.xlim(4.0, 3.4)
plt.xlabel(r'$\log(T_{\rm{eff}}/K)$')
plt.ylabel(r'$\log(L/L_{\odot})$')
```

See next page for
giant branch

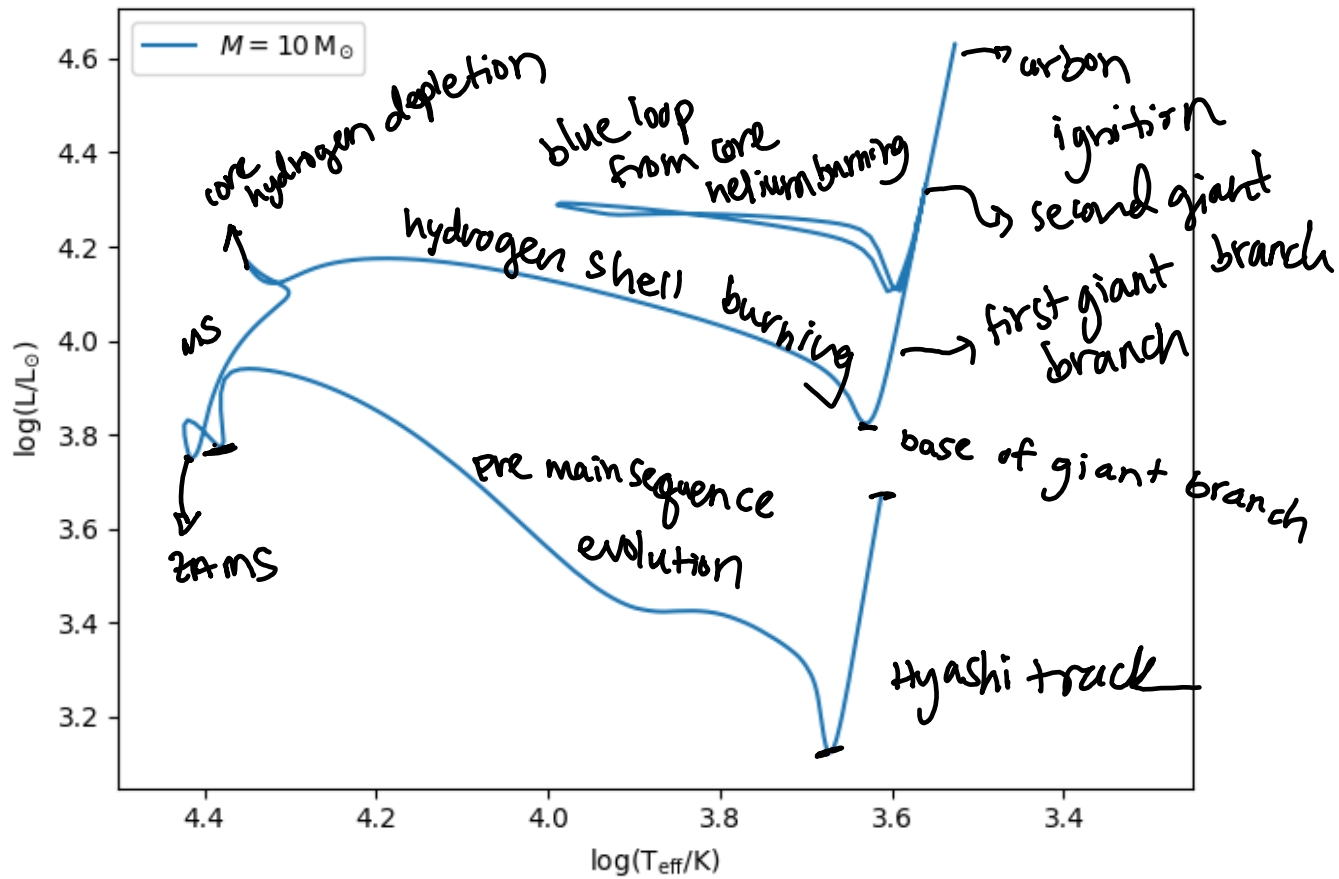
```
plt.legend()
plt.tight_layout()
```



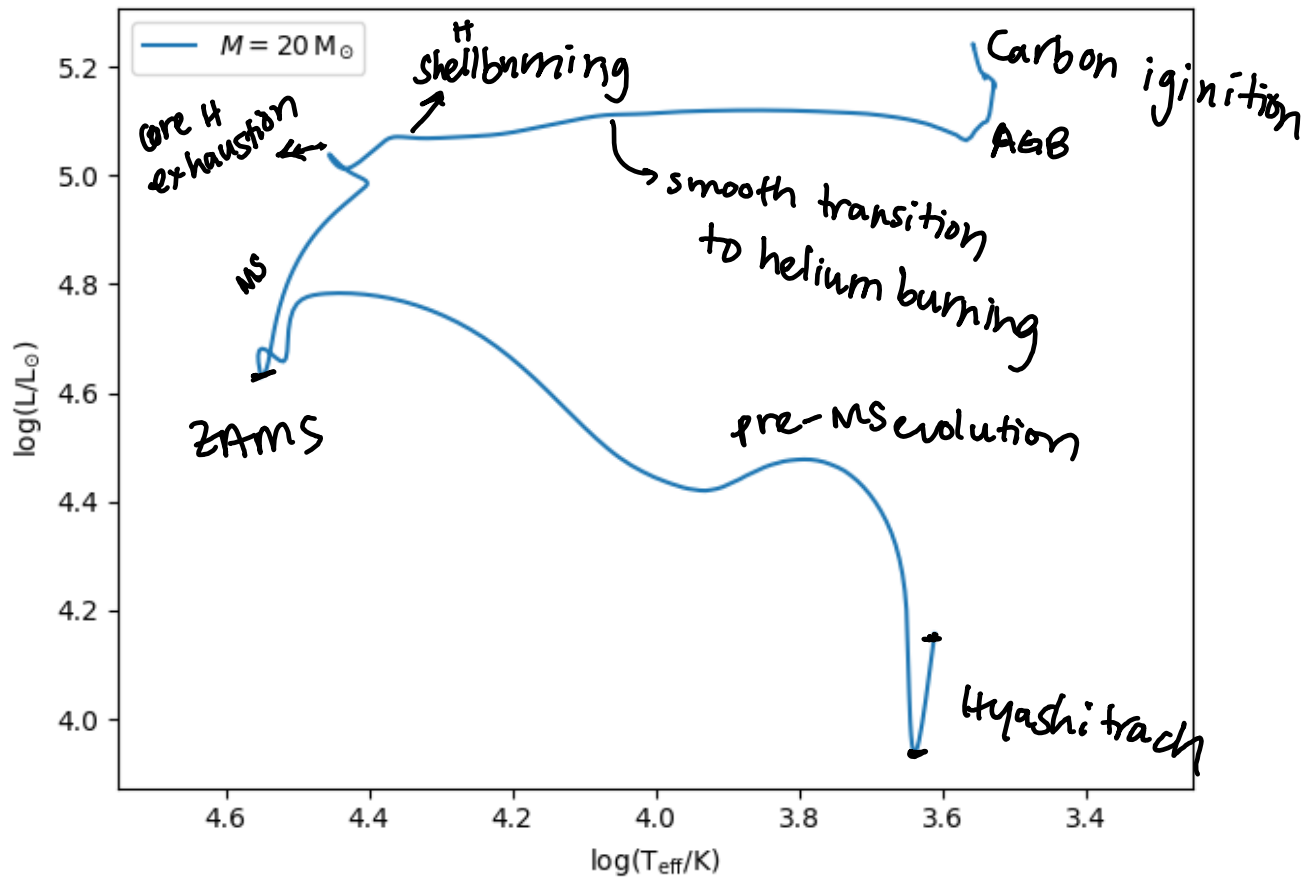
```
In [5]: plt.plot(eep5.eeps['log_Teff'], eep5.eeps['log_L'], label=r'$M=5\, \rm{M}_{\odot}$')
plt.xlim(6.0, 3.25)
plt.xlabel(r'$\log(T_{\rm{eff}}/K)$')
plt.ylabel(r'$\log(L/L_{\odot})$')
plt.legend()
plt.tight_layout()
```



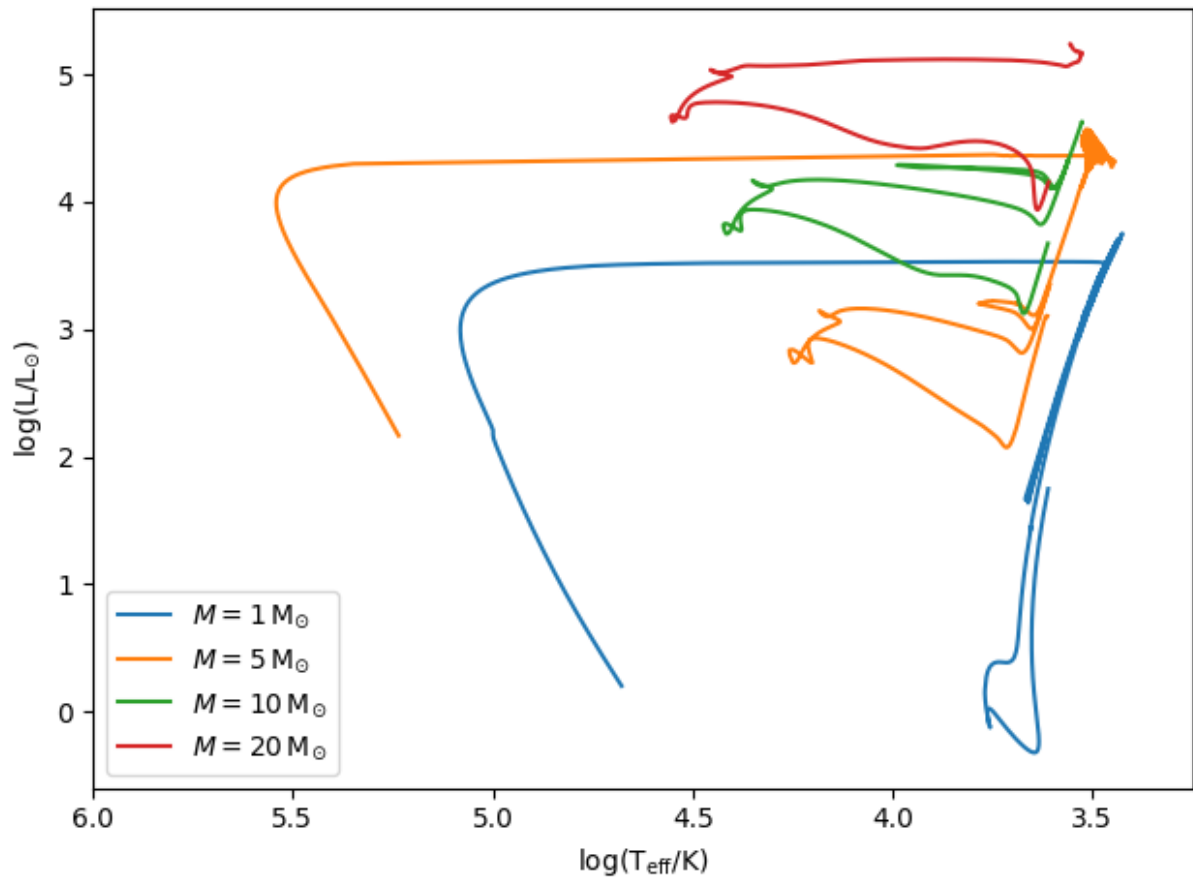
```
In [6]: plt.plot(eep10.eeps['log_Teff'], eep10.eeps['log_L'], label=r'$M=10\backslash, \rm{M}_{\odot}$')
plt.xlim(4.5, 3.25)
plt.xlabel(r'$\log(T_{\rm{eff}})/K$')
plt.ylabel(r'$\log(L/L_{\odot})$')
plt.legend()
plt.tight_layout()
```



```
In [7]: plt.plot(eep20.eeps['log_Teff'], eep10.eeps['log_L'], label=r'$M=20\, \rm{M}_{\odot}$')
plt.xlim(4.75, 3.25)
plt.xlabel(r'$\log(T_{\rm{eff}}/K)$')
plt.ylabel(r'$\log(L/L_{\odot})$')
plt.legend()
plt.tight_layout()
```



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In [8]: plt.plot(eep1.eeps['log_Teff'], eep1.eeps['log_L'], label=r'$M=1$, \rm{M}_{\odot}$')
plt.plot(eep5.eeps['log_Teff'], eep5.eeps['log_L'], label=r'$M=5$, \rm{M}_{\odot}$')
plt.plot(eep10.eeps['log_Teff'], eep10.eeps['log_L'], label=r'$M=10$, \rm{M}_{\odot}$')
plt.plot(eep20.eeps['log_Teff'], eep20.eeps['log_L'], label=r'$M=20$, \rm{M}_{\odot}$')
plt.xlim(6.0, 3.25)
plt.xlabel(r'$\log(T_{\rm{eff}}/K)$')
plt.ylabel(r'$\log(L/L_{\odot})$')
plt.legend()
plt.tight_layout()
```



In []:

#1a: 2 points per plot, 2 points for having data; -0.5 points per missing unit or label

#1b: 4 points per description
-0.25 points per missing description

#2: participation points; 1 point per "thing" written down (6pts) $\frac{1}{2}$ 1 point per strategy.